Md Farhamdur Reza

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Research Summary

Ph.D. Candidate in Electrical and Computer Engineering at North Carolina State University, specializing in **adversarial machine learning, AI safety, and decentralized federated learning**. My research develops robust, transferable, and trustworthy learning methods for vision, multimodal, and distributed AI systems. I have published first-author papers at *ICLR* and *ICCV*, introducing query-efficient geometric attacks for evaluating neural network robustness. My current work focuses on **generative transferable adversarial attacks** and **jailbreak attacks on VLMs/LLMs**, including a NeurIPS 2026 submission that analyzes the reconstruction–concealment tradeoff in multimodal jailbreaks. I also investigate alignment and post-training methods to study safety, controllability, and refusal behavior in large models. With over six years of university-level teaching experience, I combine rigorous instruction with hands-on experimentation to prepare students for the next generation of intelligent systems.

Research Interests

Adversarial Machine Learning; AI Safety and Robustness; LLM/VLM Alignment and Jailbreak Analysis; Generative and Transferable Adversarial Attacks; Decentralized Federated Learning; Computer Vision.

Education

North Carolina State University — Raleigh, NC
Doctor of Philosophy in Electrical Engineering; GPA: 4.00/4.00 — Fall 2021 - Present
Advisor: *Prof. Huaiyu Dai*

Rajshahi University of Engineering & Technology — Rajshahi, Bangladesh
Master of Science in Electrical & Electronic Engineering; GPA: 4.00/4.00 — May 2015 - Oct. 2017
Advisor: *Prof. Md Selim Hossain*

Rajshahi University of Engineering & Technology — Rajshahi, Bangladesh
Bachelor of Science in Electrical & Electronic Engineering; GPA: 3.90/4.00 (1st among 123 students) — Mar. 2010 - Dec. 2014

Research Experience

Graduate Research Assistant, Department of ECE, NC State University, Raleigh, NC — **Fall 2021 – Present**

- **CGBA (ICCV'23, Oral):** Designed a Curvature-Aware Geometric Black-Box Attack exploiting local boundary geometry, achieving state-of-the-art success across image classifiers.
- **GSBA-K (ICLR'25):** Proposed a Top-K Geometric Score-Based Black-Box Attack that efficiently manipulates multiple top predictions simultaneously using novel gradient estimators.
- **Federated Learning (ICASSP'26):** Designed mobility-aware decentralized learning frameworks that mitigate catastrophic forgetting and improve convergence speed under network constraints.
- **Generative Transferable Attack (TIFS'26, Under Review):** Proposed a generative framework for targeted adversarial transferability using pruned-ensemble discriminators and core target samples to craft adversarial examples that transfer to unseen victim models.
- **MLLM Jailbreaks (NeurIPS'26, Submitted):** Proposed *Conceal, Reconstruct, Jailbreak*, a multimodal jailbreak framework that exploits the reconstruction–concealment tradeoff to evaluate safety vulnerabilities in large multimodal models.
- **Alignment Research:** Investigating LoRA/QLoRA fine-tuning and alignment/post-training methods to study controllability, refusal behavior, and safety trade-offs in large models.
- Supervised and mentored undergraduate and graduate researchers in adversarial learning and federated optimization, with emphasis on reproducibility and ethical AI.

Selected Projects

- **Understanding Alignment Drift in Llama-3 via Multi-Stage SFT + DPO:** Designed a reproducible alignment pipeline (Helpful SFT → Safety SFT → DPO) using QLoRA/PEFT/TRL. Built 50k+ mixed safety datasets and performed controlled refusal-behavior analysis, demonstrating alignment drift under benign SFT and safety restoration via safety SFT+DPO (0.97 refusal SR).
- **Terrain Identification:** Developed CNN and time-series models to classify human activities using imbalanced accelerometer and gyroscopic data.

- **Feature Learning by Inpainting Using GAN:** Designed GAN-based models to restore missing image pixels.
- **Eye Cataract & Glaucoma Detection:** Designed a classification model using Logistic Regression, Random Forest, and SVM to identify cataract and glaucoma conditions.
- **Banknote Authentication:** Implemented ANN and Decision Tree classifiers for fraud detection.
- **Data-Driven Classification Models:** Implemented variety of classifiers for pattern recognition tasks, including linear and quadratic regression, logistic regression, k-nearest neighbors, naïve Bayes, and optimal Bayes.

Selected Publications

[1] **M. F. Reza**, A. Rahmati, T. Wu, and H. Dai, "CGBA: Curvature-aware Geometric Black-box Attack." In *Proceedings of the IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2023 (Oral).

[2] **M. F. Reza**, R. Jin, T. Wu, and H. Dai, "GSBA^K: Top-K Geometric Score-based Black-box Attack." In *Proceedings of the International Conference on Learning Representations (**ICLR**)*, 2025.

[3] **M. F. Reza**, et al., "Conceal, Reconstruct, Jailbreak: Exploiting the Reconstruction–Concealment Tradeoff in MLLMs." **Submitted** to *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2026.

[4] **M. F. Reza**, R. Jin, T. Wu, M. F. Pervej, and H. Dai, "Boosting Targeted Adversarial Transferability: A Generative Approach Using Pruned Ensembles and Core Target Samples." **Under review** in *IEEE Transactions on Information Forensics and Security (**TIFS**)*, 2026.

[5] **M. F. Reza**, et al., "Distribution-Aware Mobility-Assisted Decentralized Federated Learning." In *IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**)*, 2026.

[6] R. Jahani, **M. F. Reza**, et al., "Mobility-Assisted Decentralized Federated Learning: Convergence Analysis and A Data-Driven Approach." **Under review** in *IEEE Transactions on Cognitive Communications and Networking*, 2026.

[7] G. Li, **M. F. Reza**, et al., "Communication-Efficient and Fault-Tolerant Decentralized Federated Learning with Sparse Random Networks." **Under review** in *IEEE Transactions on Information Forensics and Security (**TIFS**)*, 2026.

[8] W. Yu, **M. F. Reza**, et al., "Curvature-Aware Geometric Black-Box Attack with Attention-Guided Dynamic Masking." **Under review** in *IEEE Transactions on Dependable and Secure Computing*, 2026.

Teaching Experience

Graduate Teaching Assistant, Department of ECE, NC State University, Raleigh, NC **Fall 2021–Present**

- Assisted in ECE 200 (*Introduction to Signals, Circuits, and Systems*), ECE 211 (*Electric Circuits*), ECE 301 (*Linear Systems*), ECE 331 (*Principles of Electrical Engineering*), and ECE 531 (*Digital Signal Processing*).
- Designed lab assignments and evaluation rubrics; mentored students in machine learning projects, emphasizing conceptual understanding and experimentation.
- Conducted weekly office hours and recitation sessions to clarify concepts, review assignments, and provide individualized feedback.
- Assisted in preparing exam questions, grading, and proctoring, ensuring fairness and transparency in evaluation.

Assistant Professor, Rajshahi University of Engineering & Technology, Bangladesh **Mar., 2018–July 2021**

- Taught undergraduate courses: EEE 3101 (*Signals and Linear Systems*), EEE 3109 (*Computational Methods in Electrical Engineering*), EEE 3203 (*Power Electronics*), and EEE 4107 (*Digital Signal Processing*), with corresponding lab class.
- Supervised 10 undergraduate theses, guided topic selection, methodology, experiments, and thesis writing; produced 5 conference papers and 1 journal.
- Served as *course coordinator* and *academic advisor*; promoted collaborative, inclusive learning practices.

Lecturer, Rajshahi University of Engineering & Technology, Bangladesh **Sep., 2015–Feb., 2018**

- Taught EEE 1201 (*Electrical Circuits II*), EEE 2205 (*Electrical Machine II*), EEE 3203 (*Power Electronics*),with corresponding lab sessions.
- Designed and maintained lab manuals emphasizing safe, hands-on experimentation and clear theoretical linkage.
- Implemented structured evaluation methods balancing theoretical mastery and practical performance

Leadership and Service

Reviewer: ICML 2026, ICLR 2026, CVPR 2025, ICLR 2025, NeurIPS 2024, PEEIACON 2024, ICECTE, ICEEE.

Conference Participation: ICCV 2023, TENSYMP 2020, EICT 2019, iCEEiCT 2018, ICECTE 2016.

Mentorship: Mentored a GEARS undergraduate researcher at NC State on a machine learning project and advised 10+ undergraduate theses at RUET.

Memberships: IEEE Graduate Student Member; Student Member of IEEE Computer Society.

Technical Skills

- **Machine Learning & AI:** Adversarial robustness, black-box attacks, transferable adversarial attacks, jailbreak evaluation, model alignment, federated/decentralized learning, representation learning.
- **Language Models & Post-Training:** Hugging Face Transformers, PEFT, TRL, LoRA/QLoRA, SFT, DPO, ORPO, RLHF, prompt engineering, safety/refusal evaluation.
- **Generative & Vision Frameworks:** PyTorch, TensorFlow, scikit-learn, OpenCV, Diffusers, GANs, diffusion models, CLIP-based evaluation.
- **Adversarial ML Tools:** RobustBench, AutoAttack, Foolbox; adversarial training and robustness evaluation pipelines.
- **Programming & Systems:** Python, CUDA, C/C++, MATLAB, Bash; Linux, Git, LaTeX, LSF/HPC.

Graduate Coursework

Taken at NC State University: Theoretical Foundation of Large-Scale Machine Learning, Pattern Recognition, Neural Networks, Optimizations and Algorithms, Introduction to Wireless Networking.

Completed at Rajshahi University of Engineering & Technology: Engineering Analysis, Information and Coding Theory, Antenna Array Signal Processing, Digital Signal Processing, Computational Electromagnetics, Antennas and Propagation.

Honors & Awards

Silver Reviewer Award, International Conference on Machine Learning (ICML)	2026
Graduate Merit Award, NC State University, Raleigh, NC	2024
Travel Grant, NC State University, Raleigh, NC	2023
University Gold Medal, Rajshahi University of Engineering & Technology, Bangladesh	2019
Prime Minister Gold Medal, University Grants Commission, Bangladesh	2017
M.Sc. Scholarship, Rajshahi University of Engineering & Technology, Bangladesh	2015–2017
Student of the Year Award, Rajshahi University of Engineering & Technology, Bangladesh	2012, 2014

References

- Dr. Huaiyu Dai — Professor, Electrical & Computer Engineering, North Carolina State University
  Email: hdai@ncsu.edu — Phone: +1-919-513-0299
- Dr. Tianfu Wu — Associate Professor, Electrical & Computer Engineering, North Carolina State University
  Email: tianfu_wu@ncsu.edu — Phone: +1-919-515-4361
- Dr. Ferdous Pervej — Assistant Professor, Electrical and Computer Engineering, Utah State University
  Email: ferdous.pervej@usu.edu — Phone: +1-435-797-9549